

Contracts, Incentives, Screening, And Moral Hazard

Contract Design Inside Employment Relationships

Special Topic 7: Labor Market Design, Contracting, and Mechanism Design – Week 3

Central question

- How should labor contracts be designed when effort, type, commitment, and multitasking are imperfectly observed?
- Week 3 moves inside the employment relationship after the worker-firm match is formed.
- The labor objects are pay, performance measures, probation, promotion, monitoring, subjective ratings, retention, and risk.
- The research standard is translation: hidden object $\rightarrow$ observed margin $\rightarrow$ institutional variation $\rightarrow$ welfare.

Principal-agent baseline

$$y = f(e, \theta, x) + \varepsilon$$

- Worker effort e is costly and imperfectly observed.
- Output y depends on effort, hidden type θ , job design x , and noise ε .
- A contract conditions pay on an imperfect signal: output, attendance, quality, rating, or promotion score.
- Stronger incentives raise returns to effort but shift risk to workers.
- The core constraints are participation and incentive compatibility.

$$\max_{a,b} \mathbb{E}[y - w(y)] \quad \text{s.t. participation and incentive compatibility.}$$

Screening and hidden type

- Hidden type is not hidden effort.
- Type can mean productivity, reliability, patience, promotability, collaboration, risk, or self-control.
- Screening devices: probation, promotion ladders, nonlinear bonuses, referral rules, benefit menus, and schedule flexibility.
- A contract change can alter who enters, who exits, and how continuing workers behave.
- Lazear's performance-pay setting is the teaching benchmark because productivity gains combine incentives and sorting.

Incentives and measured productivity

Contract object	Observable labor margin	Interpretation risk
Piece rate	output per hour, pay, quits	effort versus sorting
Team bonus	store output, peer output	free riding versus peer pressure
Manager incentive	worker allocation, output, inequality	effort versus favoritism
Goal or prize	task completion, quality, persistence	motivation versus gaming

- Output is useful, but it is not effort itself.
- Good papers ask whether output rose through effort, time, sorting, task reallocation, learning, or gaming.

Multitasking, subjectivity, and relational incentives

$$y_j = f_j(e_j) + \varepsilon_j, \quad j = 1, \dots, J.$$

- If only some tasks are measured, workers reallocate effort toward rewarded tasks.
- Subjective evaluation can recover soft information: cooperation, judgment, mentoring, quality, and reliability.
- Subjectivity also creates influence activities and evaluator-specific effort.
- Relational incentives use repeated interaction, discretionary bonuses, promotion promises, and reputation.
- The labor question is which signal is rewarded and which valuable work becomes invisible.

Commitment and worker-side demand

- Some contracts solve worker-side commitment problems, not only firm-side agency problems.
- Workers may value targets, schedules, deadlines, or penalties that help them follow through on desired effort.
- Contract take-up can reveal commitment demand, but it can also reflect risk preferences, liquidity constraints, or outside options.
- Welfare interpretation changes when the contract provides discipline valued by the worker.
- The empirical design must separate commitment from coercion and sorting.

Theory-to-applied translation

- 1 Hidden object: effort, type, commitment, evaluator favoritism, multitasking, peer effects, or relational enforcement.
- 2 Observable margin: output, quality, attendance, quits, task mix, ratings, promotions, wages, or contract take-up.
- 3 Variation: contract change, field experiment, menu, probation rule, evaluator assignment, promotion threshold, or platform policy.
- 4 Mechanism threats: incentives versus sorting, effort versus ability, quantity versus quality, output versus gaming.
- 5 Welfare object: productivity, worker risk, retention, inequality, organizational output, client welfare, or worker surplus.

Frontier methods for hidden objects

Method	Best use
Output admin data	high-frequency productivity, quality, pay, and tenure
Contract-change studies	real incentive, probation, bonus, or schedule redesigns
Randomized experiments	pay formula, framing, goals, or contract menus
Evaluator designs	favoritism, influence activities, subjective ratings
Surveys linked to admin data	commitment, beliefs, fairness, risk, self-control
Digital traces and attendance	effort timing, discipline, time use, platform activity
Structural estimation	latent effort, type, commitment, and counterfactual contracts

- Primary anchor: performance pay and productivity as a contract-change labor design.
- Challenge anchor: subjective evaluation and influence activities; optional commitment path through self-control at work.
- Reproduce: show higher output after performance pay in a synthetic teaching panel.
- Diagnose: decompose output change into within-worker incentive response and workforce-composition change.
- Transfer: redesign the workflow for subjective evaluation or commitment contracts.
- Deliverable: hidden object, margin, variation, identification strategy, welfare object, and main threat.

Takeaways

- Contracts are labor-market institutions: they allocate risk, elicit effort, reveal type, and govern retention.
- Principal-agent logic is most useful when paired with a real workplace signal and a real empirical design.
- Screening, effort incentives, multitasking, subjectivity, and relational enforcement are distinct mechanisms.
- The frontier contribution is measurement: make hidden effort, type, commitment, and evaluator behavior empirically legible.